MediTriR: A Triple-Driven Approach to Retrieval-Augmented Generation for Medical Question Answering Tasks

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***Abstract*—Advances in large language models have driven progress in medical question-answering systems, but challenges remain in accuracy and relevance, especially in complex medical settings. To address this problem, we introduce MediTriR. This approach combines knowledge graph triples with a retrieval-augmented generation framework to enhance the performance of large language models in medical multiple-choice question-answering tasks. Our approach transforms the retrieved medical knowledge into structured triples and uses these triples for accurate information retrieval. Furthermore, we dynamically generate new triples based on the properties of triples to enrich the model's reasoning ability. The integration of retrieval-augmented generation enables more accurate and contextual answers by leveraging external medical triples. MediTriR is evaluated against pure large language model approaches using the MedMCQA and Medical QA datasets, showing better performance in terms of accuracy and relevance. These results highlight the potential of our approach to improve medical decision support and education.**

***Index Terms*—Natural Language Processing, Large Language Model, Retrieval-augmented generation, Knowledge Graphs, Triples, Medical question-answering tasks**

I. Introduction

In recent years, there has been rapid progress in the application of large language models (LLMs) in natural language processing. These models, such as ChatGPT [1], have demonstrated remarkable capabilities in understanding and generating human-like text. In the medical field, applying LLMs to question-answering tasks opens up new possibilities for assisting medical professionals and patients. One specific area of interest is multiple-choice questions (MCQ) in the medical domain, where the goal is to accurately select the correct answer from a set of options based on medical literature.

However, LLMs can hallucinate [2] and generate incorrect answers, which is a critical issue in some fields, such as medicine. Furthermore, data in the pre-training phase of LLM often lacks the depth required for specialized domain question-answering tasks. To address these limitations, retrieval-augmented generation (RAG) [3] techniques have been developed. RAG enhances LLM performance by combining retrieval and generation processes: it first retrieves relevant information from a large document set or database and then uses this data to inform the generation phase where the model creates responses. This integration helps improve the accuracy and relevance of answers, making RAG particularly valuable in complex fields such as medicine.

Despite the progress of LLMs in medical tasks, there are still some challenges in using LLMs for medical MCQ tasks. The main problem is to efficiently retrieve and integrate relevant information from large biomedical datasets to answer questions accurately. Traditional retrieval methods sometimes fail to capture the subtle and complex relationships in medical texts, resulting in incomplete or incorrect answers. This problem is worth studying because improving the accuracy and reliability of medical question-answering systems will have a direct impact on medical decision-making and patient outcomes.

Our research aims to develop a method, MediTriR, to improve the performance of LLM in medical MCQ tasks by leveraging knowledge graph triples. Knowledge graphs represent information in a structured format that represents the relationships between entities. By converting medical literature into knowledge graph triples and integrating them into the question-answering process, we aim to improve the model's ability to retrieve and reason about relevant information. An example of our method is shown in Fig. 1.

The contributions in this paper are as follows:

- We propose a triple-based RAG method to solve the multiple-choice question-answering task in the medical field. Replacing text with triples allows LLMs to obtain more effective information.
- We experimented with multiple models on multiple datasets and performed ablation experiments to demonstrate the effectiveness of our proposed method.
- We analyze each step of our method in detail and explain how these steps help LLMs solve MCQs in the medical field.

II. Literature review

A. Advances in RAG for Medical Question Answering

Recent advancements in retrieval-augmented language models for medical question-answering emphasize precision,

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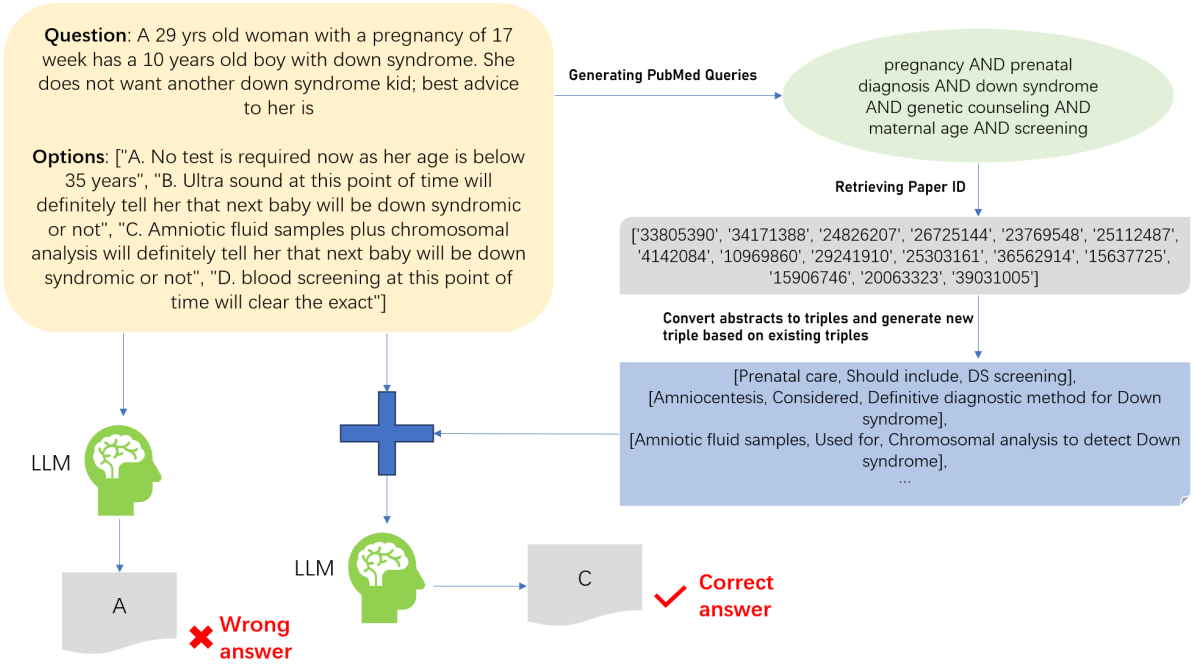


Fig. 1. An example of MediTriR.(Asking the LLM medical questions directly resulted in the wrong answer. The LLM combined with the MediTriR approach resulted in the correct answer.)

domain-specific knowledge integration, and computational efficiency. Both Clinfo.ai [4] and JMLR [5] focus on enhancing accuracy through retrieval mechanisms; Clinfo.ai synthesizes answers from summarized articles, while JMLR jointly trains language and retrieval models to reduce hallucinations and resource use. Med-PaLM 2 [6] and GatorTron [14] improve base models through fine-tuning with medical texts; Med-PaLM 2 uses ensemble refinement for high accuracy, whereas GatorTron broadens its utility across clinical tasks. ChatDoctor [17] leverages real-time retrieval to handle diverse medical inquiries effectively. These studies show that by combining expertise with efficient retrieval, the model can perform better in medical question-answering tasks.

B. Applications of Knowledge Graphs in Medical Question Answering

Zhixue Jiang [10] and Xiaofeng Huang [11] both use knowledge graphs to boost question-answering accuracy, with Jiang focusing on practical medical queries through rule-based methods and Huang enhancing disease diagnosis with weighted path rankings. Ziqing Liu [12] integrates a Bi-LSTM+CRF algorithm with a Traditional Chinese Medicine knowledge graph for precise dialogues, while Quan Guo [13] combines Elasticsearch and pre-trained models for improved semantic matching in multilingual contexts. These approaches highlight a trend towards more responsive and specialized medical question-answering systems through the integration of structured knowledge and advanced processing techniques.

C. Evaluation and Benchmarking of Medical Question Answering Systems

MEDQA [7] is a multilingual dataset from medical exams that challenges AI with complex reasoning. GreaseLM [8] combined pretrained language models with graph neural networks, significantly improving performance on medical benchmarks. Valentin Liévin [9] demonstrated that LLMs like InstructGPT can effectively answer medical exam questions. Harsha Nori [15] showed that GPT-4 outperforms previous models in medical exams, suggesting its potential for clinical practice. Zhongjing [16] is a Chinese medical LLM proficient in multi-turn dialogues. Yee Hui Yeo [18] evaluated ChatGPT's accuracy on cirrhosis and HCC, highlighting its usefulness and limitations as an educational tool.

D. Gap Analysis

Despite significant advancements in medical question-answering systems, several gaps remain in leveraging knowledge graph triples effectively. Current approaches using knowledge graphs primarily focus on enhancing question-answering accuracy through structured information integration (Zhixue Jiang et al., [10]; Xiaofeng Huang et al., [11]; Ziqing Liu et al., [12]; Quan Guo et al., [13]) or improve performance by fusing knowledge graphs with other methods (Xikun Zhang et al., [8]). Meanwhile, retrieval-augmented generation (RAG) methods have shown promise in improving the retrieval and synthesis of information for medical questions (Alejandro Lozano et al., [4]; Junda Wang et al., [5]; Karan Singhal et al., [6]; Xi Yang et al., [14]; Yunxiang Li et al., [17]), but

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often fall short in effectively incorporating domain-specific knowledge from structured data sources. Furthermore, while recent evaluations and benchmarks (Di Jin et al., [7]; Valentin Liévin et al., [9]; Songhua Yang et al., [16]; Harsha Nori et al., [15]; Yee Hui Yeo et al., [18]) provide insights into model performance, they do not fully address the integration of knowledge graph triples in the context of LLMs. Our research addresses these gaps by proposing a novel method that enhances the performance of LLMs in MCQ tasks through the systematic conversion of medical literature into knowledge graph triples. By integrating these triples into the question-answering process, our method aims to improve the retrieval and reasoning capabilities of LLMs, thus providing more accurate and relevant answers based on structured domain knowledge.

III. System Design

A. Overview of the Approach

Our approach combines structured knowledge with the generative power of LLMs. The system aims to improve the accuracy and relevance of answers generated by the model, ensuring that they are based on reliable medical literature. The process begins by generating PubMed [19] queries based on medical questions. These queries are designed to retrieve articles that contain relevant information needed to answer the questions. We then searched and retrieved paper abstracts from the PubMed database using the Entrez API [19]. In this step, we obtained rich medical information, which provided a knowledge base for the RAG process. After retrieving the abstracts, we converted them into triples in the knowledge graph. Each triple represents a piece of information in the format of (subject, predicate, object), which captures the basic relations and entities described in the abstract. This structured representation allows for more efficient and accurate processing of information. The next step involves evaluating the relevance of these triples to the medical question. This filtering process ensures that only relevant triples are retained, reducing noise and focusing the model's attention on the most relevant data. Next, we generate new triples based on the existing triples. This step exploits the extensibility of triples to infer additional useful information that may not be explicitly stated in the original abstract. Finally, the system combines the retained triples and the newly generated triples to answer the medical MCQ. The framework of our method is shown in Fig. 2.

B. Generating PubMed Queries

Inspired by [4], the first step in our approach involves transforming medical questions into PubMed search queries and retrieving relevant paper abstracts. In this step, we collected comprehensive biomedical documents to provide the basic information required for subsequent processing. We used the LLM, GPT-4o [20] to generate effective PubMed queries from medical questions. The main goal is to create queries to retrieve articles containing relevant information. For each query, we retrieved 15 papers. Such a retrieval method can maximize capturing all relevant information required to answer the question accurately. An example is shown in Fig. 3.

C. Retrieving Paper Abstracts

After generating a PubMed query, the next step is to use the Entrez API to search and retrieve relevant article abstracts from the PubMed database. The search process is performed as follows:

- Search PubMed Database: Use the generated query to search the PubMed database via the Entrez API. The search is configured to retrieve 15 relevant articles.
- Retrieve Article IDs: Extract the unique identifiers (IDs) of the retrieved articles from the search results.
- Fetch Abstracts: Use the article IDs to fetch the corresponding paper abstracts from PubMed.

D. Converting Abstracts into Triples

After retrieving the relevant paper abstracts, the next step involves converting these abstracts into knowledge graph triples and assessing their relevance to the medical question.

The conversion of paper abstracts into knowledge graph triples using an LLM, GPT-4o [20]. Each triple captures the core relationships and entities described in the abstract. This structured representation allows for the more efficient processing and reasoning of biomedical information.

We instruct an LLM, GPT-4o [20], to act as an assistant that converts abstracts into triples. The model is given the task of reading the abstracts and extracting information in the form of triples. For instance, an abstract describing a medical trial might be converted into triples like (Drug A, Treats, Disease B) or (Disease B, Associated With, Symptom C). These triples encapsulate the key pieces of information in a structured format that can be easily queried and analyzed. The prompt is shown in Fig. 4.

E. Assessing Relevance of Triples

Once the triples are generated, the next step is to filter them, keeping only those relevant to the medical question. This step ensures that the model focuses on the most relevant information, thereby reducing noise and improving the accuracy of the answer.

Relevance assessment is performed by instructing the LLM to act as a medical researcher, evaluating whether each triple contains information that may be relevant to the medical question. The model reviews each triple in the context of the question and keeps only those considered relevant. For example, if the medical question is about the treatment of a specific disease, only triples related to the treatment, medication, or medical outcome of that disease will be kept.

By combining these steps, we ensure that our system has a structured and filtered knowledge graph that is triple-relevant to the medical question.

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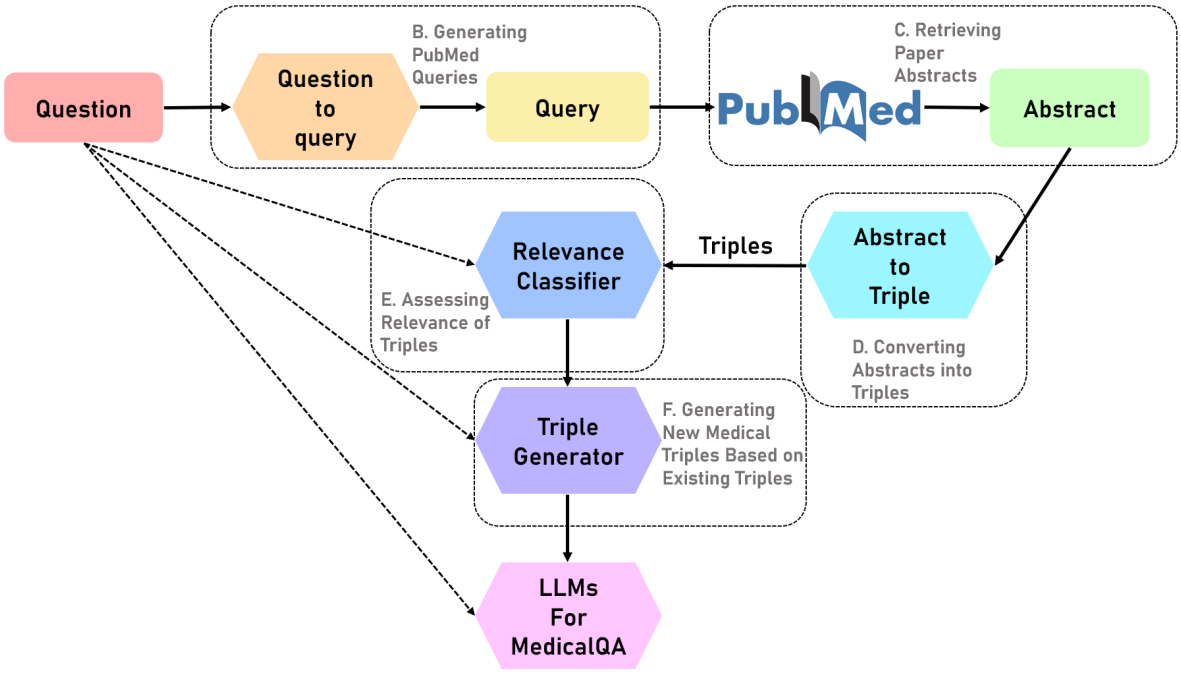


Fig. 2. The framework of MediTriR.

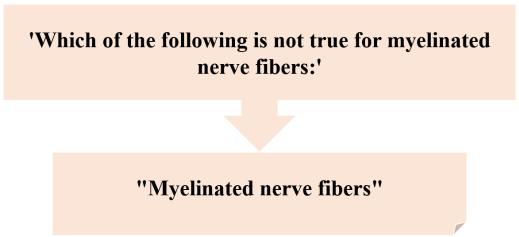


Fig. 3. An example of generating PubMed queries.

```
text2tri_sys = "You are a helpful assistant that converts
abstracts of PubMed articles into triples that can be used to
answer clinical questions. Each triple should follow the format:
(Subject, Predicate, Object)."
text2tri_prompt = "Convert the following abstract into triples
that can be used to answer the clinical question:
\"{question}\":\n\nAbstract: \"\"\"{article_text}\"\"\"\n\nEach
triple should follow the format: (Subject, Predicate, Object)."
```

Fig. 4. Prompt for converting abstracts into triples.

F. Generating New Medical Triples Based on Existing Triples

After converting the abstracts into knowledge graph triples and filtering for relevance, the next step is to use the LLM to generate new medical triples based on existing triples. This step exploits the extensibility of triples to infer additional useful information that may not be explicitly stated in the original abstract.

We use the LLM to generate new triples. The goal is to create new accurate triples by logically combining or inferring information from a given set of existing triples. This process helps to expand the knowledge base and discover implicit relationships that are not directly mentioned in the retrieved abstracts but are valuable for answering medical questions.

We instruct the LLM to act as an assistant to generate new triples from a list of existing triples. The model's task is to review existing triples and create new triples that follow the same format. At the same time, we also pass in the original medical question as a reference for the LLM when generating new triples. These new triples should be meaningful and error-free, ensuring that they make positive contributions to the knowledge base.

For example, if the existing triples include (drug A, treatment, disease B) and (disease B, related, symptom C), the model may generate a new triple such as (drug A, reduction, symptom C). This new triple follows logically from existing information and provides additional insights that may be valuable in answering medical questions. The prompt is shown in Fig. 5.

IV. Experiments and Results

In this chapter, we will describe our experiments in detail. First, we will introduce the data set and LLMs we used, followed by our experimental results.

A. Dataset

In our experiments, we utilized two primary datasets from the official United States Medical Licensing Examination

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```
generate_sys = """
You are a helpful assistant that generates new triples from a list of existing
triples. Each triple follows the format: (Subject, Predicate, Object).
Your task is to generate new, accurate triples by logically combining or
inferring information from the given triples.
Ensure that the new triples are meaningful to the medical questions and do not
contain any errors.
"""
generate_prompt = """
Generate new triples for medical question-answering problem from the following
list of existing triples:
Question:
{question}
Answer Options:
{option}
Existing Triples:
{existing_triples}
New Triples:
Only return the relevant medical triples with no explanation.
"""
```

Fig. 5. Prompt for generating new medical triples based on existing triples.

(USMLE) website to evaluate the performance of our approach: MedMCQA and Medical QA. These datasets provide a comprehensive set of medical MCQs, which are essential for assessing the accuracy and effectiveness of our method.

1) MedMCQA Dataset [21]: The MedMCQA dataset is a widely used benchmark for medical question-answering tasks. It consists of MCQs covering a broad range of medical topics, designed to test the knowledge and reasoning capabilities of models in the medical domain. This dataset includes questions that require an in-depth understanding of medical concepts and the ability to apply this knowledge to specific medical scenarios.

For our experiments, we focused on the dev set of the MedMCQA dataset. The dev set of MedMCQA has 4183 questions.

2) Medical QA Dataset [7]: The Medical QA dataset from the official USMLE website is another crucial resource for our experiments. This dataset provides a high standard of medical knowledge assessment. The questions are crafted to evaluate a wide range of medical knowledge and medical skills, ensuring that our method is tested against real-world medical scenarios.

The Medical QA dataset is from three regions: the United States, Mainland China, and the Taiwan District. For our experiments, we utilized the US dev subset of the Medical QA dataset. This subset contains a diverse set of questions that reflect the medical knowledge required for practicing medicine in the United States. By focusing on the US dev dataset, we ensure that our evaluation is aligned with the standards and expectations of medical professionals in the US. The dev set of the Medical QA dataset has 1272 questions.

B. Baseline Models

In our experiments, we evaluated the performance of our approach against several baseline models. These models include state-of-the-art LLMs known for their capabilities in text generation, question answering, and reasoning. The models we used are gemma2-9b-it, llama-3-8b, ChatGPT-3.5, and GPT-4o mini. Below, we provide a brief overview of each model.

1) Gemma [22] (gemma2-9b-it): The gemma2-9b-it model is part of Google's Gemma family of lightweight, state-of-the-art open models. The Gemma model was released in June 2024 and is built using the same technology as the Gemini model. The Gemma model is a text-to-text, decoder-only LLM available in English with open weights, pre-trained, and instruction-tuned variants.

Gemma performs well on text generation problems such as reasoning, summarization, and question-answering. In terms of resource usage, the Gemma model has a lower resource usage than most LLMs. This makes it particularly suitable in resource-constrained environments, such as laptops, desktops, or private cloud infrastructure.

2) Llama3 [23] (llama-3-8b): The llama-3-8b model is another prominent LLM used in our experiments. Known for its robust performance in various natural language processing tasks, Llama-3-8b has been widely adopted in the research community. This model excels in text generation, comprehension, and reasoning tasks, making it a strong candidate for medical question-answering applications.

Llama-3-8b's architecture and training data enable it to understand complex medical terminology and concepts, providing accurate and contextually appropriate responses to medical questions.

3) ChatGPT [1] (ChatGPT3.5, GPT-4o mini): The ChatGPT-3.5 model is a well-known LLM developed by OpenAI. It has been extensively used for conversational AI applications and demonstrates strong performance in generating human-like text. ChatGPT-3.5 is built on the GPT-3.5 architecture, which leverages a massive dataset and advanced training techniques to understand and generate text across a wide range of topics.

In the context of answering medical questions, ChatGPT-3.5 can interpret and respond to complex queries with a high degree of accuracy.

C. Main Results

In this section, we present the results of our MediTriR method applied to two medical MCQ datasets, MedMCQA [21] and Medical QA [7], comparing its performance against traditional LLMs without the MediTriR enhancements. The results are shown in Table 1 and Fig. 6.

TABLE I
EXPERIMENTAL RESULTS ON TWO MEDICAL DATASETS.

Model	MedMCQA (%)	Medical QA (%)
gemma2-9b-it	54.12	47.09
gemma2-9b-it + MediTriR	63.83	56.05
llama3-8b	47.90	43.63
llama3-8b + MediTriR	61.53	47.64
gpt-3.5-turbo	51.06	34.75
gpt-3.5-turbo + MediTriR	66.93	43.24
gpt-4o-mini	65.96	63.68
gpt-4o-mini + MediTriR	74.32	70.28

Our results demonstrate that integrating the MediTriR method improves the performance of all tested models on both datasets. To illustrate the impact of MediTriR, consider the performance enhancements observed with the llama3-8b model. Initially, llama3-8b scored 47.90 on MedMCQA and 43.63 on Medical QA. With the integration of MediTriR, these

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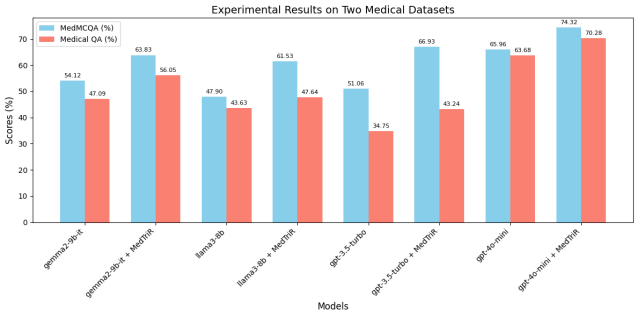


Fig. 6. Experimental results on two medical datasets.

scores improved dramatically to 61.53 and 47.64, respectively. Similarly, the gpt-4o-mini model, already performing well at 65.96 on MedMCQA and 63.68 on Medical QA, saw its scores increase to 74.32 and 70.28 with MediTriR. The integration of MediTriR consistently improves the performance of various LLMs on medical MCQ tasks. This improvement is reflected in the increase in accuracy and contextual relevance of all tested models. MediTriR structures input data as knowledge graphs triple and enhances the generation process with triple-based retrieval, enabling models to parse and utilize complex medical information.

D. Ablation Experiment

In this section, we evaluate the impact of generating new triples on the performance of our MediTriR method. This ablation study aims to illustrate the overall effectiveness of the triple generation step on our method for the medical multiple-choice question-answering task.

Our ablation experiments compare the performance of models augmented with the MediTriR approach with and without the new triple-generation step. This step involves using existing knowledge graph triples to infer and generate additional triples that are not explicitly stated in the retrieved data but may be relevant to the medical question. The experiments are conducted on two datasets: MedMCQA and Medical QA. The results are shown in Table 2, Table 3, Fig. 7 and Fig. 8.

TABLE II
RESULTS FOR ABLATION EXPERIMENT ON MEDMCQA.

Model	No triple generation (%)	MediTriR (%)
gemma2-9b-it	61.30	63.83
llama3-8b	57.76	61.53
gpt-3.5-turbo	64.50	66.93
gpt-4o-mini	70.67	74.32

From the results in the two tables above, we can find that including the step of generating new triples can improve the performance of the model on both data sets. For example, on the MedMCQA dataset, the performance improvement of the gpt-4o-mini model is very obvious, with the score improving from 70.67 to 74.32 after generating new triples. Similar

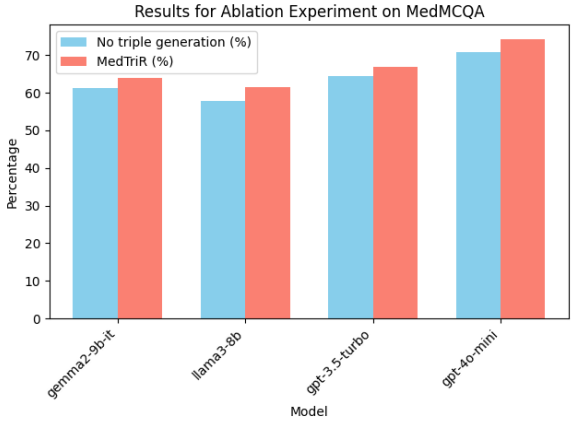


Fig. 7. Results for ablation experiment on MedMCQA.

TABLE III
RESULTS FOR ABLATION EXPERIMENT ON MEDICAL QA.

Model	No triple generation (%)	MediTriR (%)
gemma2-9b-it	53.46	56.05
llama3-8b	45.83	47.64
gpt-3.5-turbo	40.96	43.24
gpt-4o-mini	67.61	70.28

improvements were observed in the Medical QA dataset, where the performance of gpt-4o-mini improved from 67.61 to 70.28.

These improvements demonstrate that generating new triples enables the model to achieve a deeper understanding and broader coverage of the medical topic in question. By inferring new relationships and expanding the knowledge base, models can better contextualize and answer complex medical questions. This ability to generate and utilize new triples effectively addresses gaps in the retrieved data, providing more

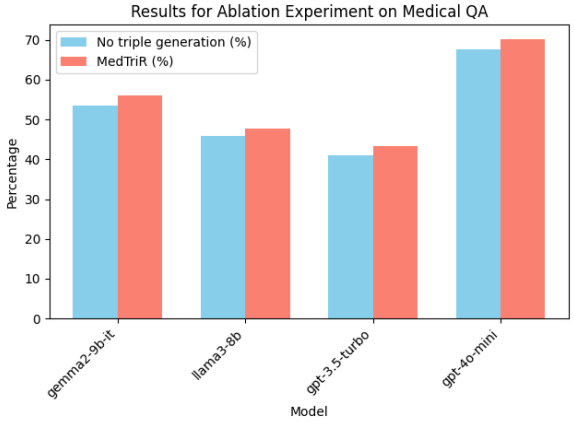


Fig. 8. Results for ablation experiment on Medical QA.

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comprehensive and accurate answers.

Ablation studies highlight the value of generating new triples in the MediTriR approach. This step improves our approach's overall performance by extending the information base explicitly provided in the source documentation.

V. Discussion

In this chapter, we discuss the advantages of our experimental approach to the medical multiple-choice question-answering task. Our approach leverages the RAG approach and is enhanced using knowledge graph triples. This innovative combination provides multiple benefits that can improve the accuracy and reliability of medical question-answering systems.

A. Benefits of the RAG Method for Medical Problems

The first step in our approach involves converting medical questions into search queries suitable for retrieving relevant medical literature. This is accomplished using an LLM specifically instructed to generate PubMed queries based on the given questions.

By converting medical questions into well-formulated search queries, we can target the retrieval of the most relevant documents from PubMed. This ensures that the retrieved information is closely aligned with the specific needs of the question, reducing the likelihood of irrelevant data being included in the subsequent steps.

Once the queries are generated, the next step involves using the Entrez API to search for paper abstracts in PubMed. The advantages of this approach include:

- PubMed is one of the most extensive medical literature databases, containing millions of abstracts from peer-reviewed journals. By leveraging PubMed, we ensure that the information retrieved comes from authoritative and recent research articles, which is critical to maintaining accuracy and trust in answers.
- The Entrez API allows precise control of search parameters, thereby retrieving the documents most relevant to the query. This targeted retrieval reduces noise in the data and ensures that subsequent steps in our approach use the highest quality information.
- The retrieved abstracts provide a rich source of information that LLM can use to generate accurate answers. By combining this external data with the model's existing knowledge, we enhance the model's ability to effectively answer complex medical questions.

The strength of the RAG approach is its ability to combine retrieving information and generating answers. A common problem with generative models is hallucinations, producing information that appears plausible but is incorrect. The RAG approach reduces the risk of hallucinations by first retrieving relevant documents and then using those documents to generate answers. This combination ensures that answers are not only coherent and contextual but also grounded in validated medical literature.

B. Benefits of Converting Text Data into Triples

1) Enhanced Semantic Information: Triples provide a clear and structured representation of the information contained in the abstract. Such a structured format makes it easier for the LLM to process and understand the relationships between different medical concepts. By breaking down complex sentences into simpler, more understandable units, the LLM can effectively use this information to generate accurate answers.

By converting abstracts into triples, we extract the most relevant information from medical papers. This extraction process filters out irrelevant details, allowing the model to focus on key facts and relationships directly related to the medical question. This focus enhances the relevance and quality of the information used by the model, resulting in more accurate answers.

2) Accurate Relevance Assessment: By evaluating the relevance of triples, we try to ensure that LLM uses only the most relevant information in the answer-generation process. Precision knowledge and information are very important in the medical field, as accurate and reliable information is important for patient care. By filtering out irrelevant triples, we reduce the risk of incorporating misleading or irrelevant information into the answer.

Evaluating the relevance of triples simplifies the data processing workflow. By focusing only on relevant triples, we can minimize the computational resources required for subsequent steps. This efficiency also helps in real-time applications that require fast and accurate responses.

3) Scalability introduces more knowledge: Generating new triples allows us to expand the knowledge base beyond the information initially retrieved. By logically combining existing triples, the model can generate new, meaningful triples that provide additional insights. For example, if we have a triple stating that drug A can treat disease B and another triple stating that disease B is associated with symptom D, the model may generate a new triple stating that drug A can alleviate symptom D. This expansion based on the unique properties of triples can bring more valuable insights to the medical field.

Generating new triples enhances the model's ability to provide accurate and detailed answers. By supplementing the original information with inferred data, the model can combine knowledge from multiple papers to provide more comprehensive answers and address the nuances of medical questions. This capability is particularly useful in complex medical scenarios where multiple factors need to be considered.

Not all relevant information is always explicitly stated in the medical literature. By generating new triples, the model can fill in the gaps in the existing knowledge base, ensuring that all necessary information is available to answer the question. This proactive approach mitigates the limitations of incomplete data and improves the overall robustness of the system.

VI. Conclusion

In this paper, we propose an innovative approach, MediTriR, to improve the accuracy and relevance of answers provided

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by LLMs in medical multiple-choice tasks. Our approach combines the strengths of the RAG framework with a novel triple-based data representation and shows great potential in improving the quality of medical multiple-choice tasks.

Our approach first uses LLM to convert medical questions into precise PubMed search queries, ensuring the retrieval of highly relevant medical literature. This initial step leverages PubMed's extensive resources and the precision of the Entrez API, which together facilitate the extraction of authoritative and up-to-date information. This process addresses the need for reliable and relevant data sources.

After retrieving paper abstracts, we convert these abstracts into structured triples, which provide a clear and manageable format for representing complex medical information. This step enhances the clarity and relevance of the data, allowing the LLM to focus on the essential relationships between medical concepts. By evaluating the relevance of these triples to the medical question, we ensure that only the most relevant information is used, further improving the accuracy of the generated answers.

One key innovation of our approach is generating new triples based on existing triples. This feature allows the system to expand its knowledge base and infer more relevant information, thereby filling gaps and improving the comprehensiveness of answers. Such innovations can be helpful in medical tasks, as the medical field often requires complex reasoning and comprehensive knowledge.

Our approach can reduce the risk of hallucination problems common in generative models by basing the generation process on retrieved, verified information. This foundation ensures that the answers are coherent, contextual, and based on reliable medical literature.

Although our approach has shown promising results, several areas still require future research and development. Further refinement of the query transformation process and relevance evaluation algorithm can improve the accuracy and efficiency of our approach. In addition, expanding the scope of medical databases beyond PubMed and integrating real-time updates of ongoing medical research can further improve the accuracy and relevance of the system.

In summary, our triple-based RAG approach shows promising results in medical MCQ tasks. By combining the advantages of structured data representation with advanced retrieval and generation techniques, our approach effectively solves the challenges of medical multiple-choice tasks.

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